Arista Access Node BAS-ACC-78 Basavanagudi POP QoS Congestion Event Resolution SOP

Node: BAS-ACC-78 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_5255

Description

The failure domain of QoS congestion events, manifesting as microbursts, interface tail drops, and QoS queue starvation on Arista's Access Node BAS-ACC-78 at the Basavanagudi POP, can be attributed to the intricate interplay between the node's role as an access layer device and the vendor's equipment-specific implementation of Quality of Service (QoS) protocols. The Arista EOS (Extensible Operating System) relies on the concept of CoS (Class of Service) and DSCP (Differentiated Services Code Point) marking to prioritize traffic and manage congestion. However, in the event of a microburst, where a large amount of traffic is transmitted in a short period, the interface queues can become overwhelmed, leading to tail drops and QoS queue starvation. This scenario is further exacerbated by the node's configuration, wherein the Access Control List (ACL) and QoS policies are tightly coupled, allowing for a small number of high-priority traffic flows to dominate the available bandwidth and cause congestion. Furthermore, the Basavanagudi POP's geographical location, with its high population density and corresponding network traffic demands, contributes to the likelihood of such congestion events. Therefore, this SOP aims to provide a step-by-step guide for troubleshooting and resolving QoS congestion events on Arista's Access Node BAS-ACC-78 at the Basavanagudi POP.

Pre-requisite Checks

- Pre-requisite: Verify that remote hands have been dispatched to the Basavanagudi POP to facilitate physical access to the node in case of a hardware issue., Description: Ensure that a team of skilled engineers is available to assist with the troubleshooting and resolution process.

- Pre-requisite: Check the status of redundant links to determine if the failure is isolated to a single link or if it is a widespread issue across multiple links., Description: Verify the configuration and operational status of all redundant links to identify potential causes of the congestion event.

Potential Root Causes

Cause	Probability
Microburst caused by a small number of high-priority traffic flows dominating the available bandwidth.	High
Inadequate QoS policies resulting in QoS queue starvation.	Medium
Interface queue overflow due to excessive traffic.	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos police, Expected Output: QoS policers: 10, Description: This command displays the list of QoS policers configured on the node. The output will help identify if any policers are misconfigured or missing, leading to QoS queue starvation.
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Command: show qos police detail, Expected Output: QoS policer 1: 10000
packets/s, Description: This command provides detailed information
about each QoS policer, including the packet rate and burst size. The
output will help identify if any policers are under- or
over-configured, contributing to congestion events.

Command: show interface statistics, Expected Output: Interface stats:
100000 packets/s, Description: This command displays the interface
statistics, including the packet rate and drop count. The output will
help identify if any interfaces are experiencing high packet rates or
drop counts, indicating potential congestion issues.
```

Resolution Steps

Step 1: Step: Configure QoS policies to prioritize traffic flows., Command: config qos class-map CL-LOW-PRIO match protocol ip dscp 0, Description: This step involves configuring a new QoS class map to prioritize low-priority traffic flows. The command creates a new class map named CL-LOW-PRIO that matches traffic with a DSCP value of 0.

Step 2: Step: Apply QoS policies to interfaces., Command: config qos interface eth1 class-map CL-LOW-PRIO, Description: This step involves applying the QoS policies to the interface eth1. The command applies the QoS class map CL-LOW-PRIO to the interface eth1.

Step 3: Step: Verify QoS policies., Command: show qos interface, Description: This step involves verifying the QoS policies applied to the interface. The command displays the QoS configuration for the interface, including the class map and police actions.

Post-Check Verification

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Command: show qos interface, Expected Output: QoS config: Class map
CL-LOW-PRIO applied to interface eth1, Description: This command
verifies the QoS configuration applied to the interface. The output
will confirm if the QoS class map is applied correctly.

Command: show interface statistics, Expected Output: Interface stats:
50000 packets/s, Description: This command verifies the interface
statistics after applying the QoS policies. The output will confirm if
the congestion event has been mitigated.
```

Roll-back Procedures

- Step: Remove QoS policies., Command: config no qos interface eth1 class-map CL-LOW-PRIO, Description: This step involves removing the QoS policies applied to the interface. The command removes the QoS class map CL-LOW-PRIO from the interface eth1.

- Step: Revert to previous QoS configuration., Command: config restore qos, Description: This step involves reverting to the previous QoS configuration. The command restores the previous QoS configuration from the configuration database.